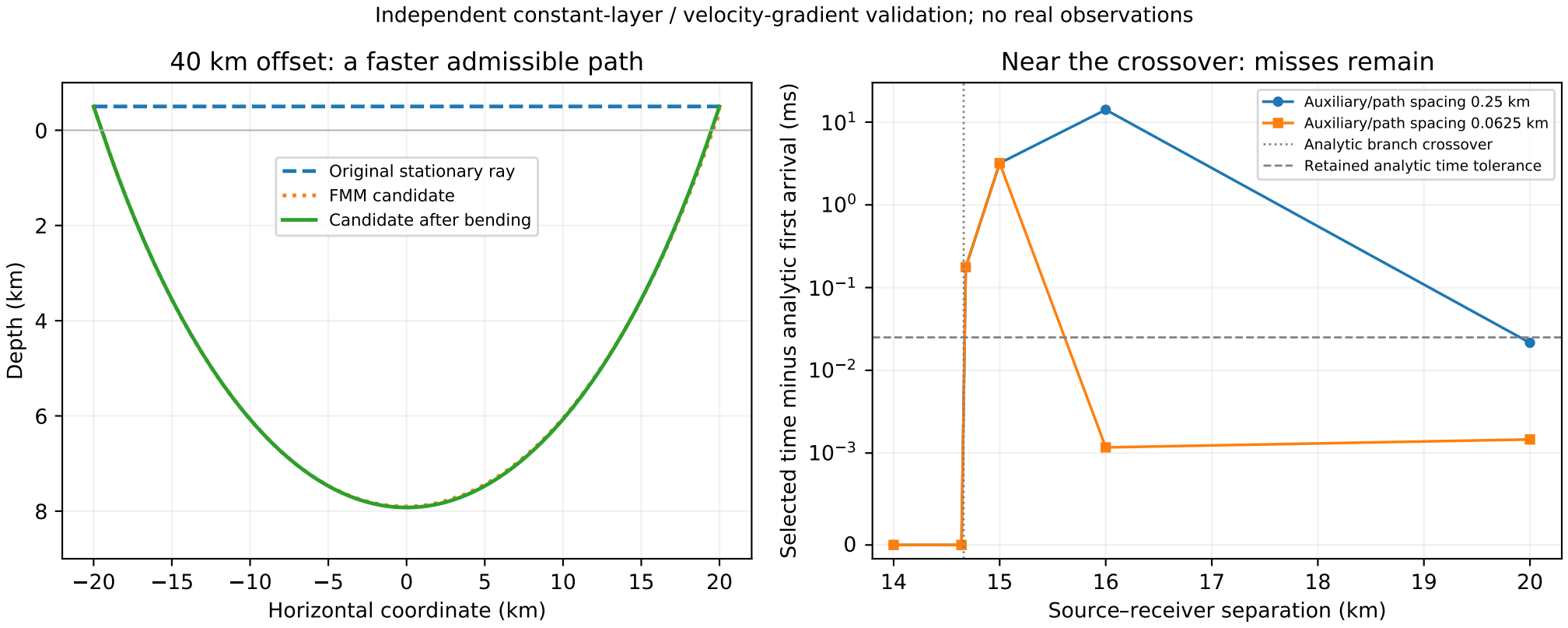
Independent constant-layer / velocity-gradient validation; no real observations
40 km offset: a faster admissible path
Original stationary ray
FMM candidate
Candidate after bending
Depth (km)
Horizontal coordinate (km)
Near the crossover: misses remain
Auxiliary/path spacing 0.25 km
Auxiliary/path spacing 0.0625 km
Analytic branch crossover
Retained analytic time tolerance
Selected time minus analytic first arrival (ms)
Source–receiver separation (km)